Probability and Statistics for Beginners

Topic	Module	Lessons	Description of lesson
Probability (Beginners)	1.Introduction to Probability	1. What is Probability 2. Event Outcomes	1. Know about probability definitions 2.
	2. Basic Set Operations	1. Basic Sets 2. Union and intersection 3. Disjoin set 4. Complement Sets 5. Properties Of Set Operations	
	3. Set Theory and Models	1. Tree Diagram 2. Probability Example 3. Addition Rule 4. Multiplication Rule	
	4. Experimental Probability VS Theoretical Probability	1. Definition and Comparison 2. Experimental Probability 3. Theoretical Probability	
	5. Randomness	1. What is Randomness? 2. Importance of Randomness 3. Simulating Randomness	
	6. Mutual and Non-mutual exclusive events.	1. Mutual and Non-Mutually Exclusive Events 2. General Addition Rule	

	7. Independent & Dependent Events	1. Independent vs Dependent Events 2. Multiplication Rule for Independent Events 3. Multiplication Rule for Dependent Events	

Introduction to Descriptive Statistics	1. Descriptive Statistics 2. Mean, median, mode and range 3. Range, Maximum, Minimum, and Standard Deviation
Introduction to Inferential Statistics	1. Inferential statistics 2. Population & Sample 3. Hypothesis testing 4. Regression analysis

Probability and Statistic for Intermediate

Topic	Module	Lessons
Probability (Intermediate)	1. Recap to: Probability	1. Sample Space and Events 2. Union (Additional Rule) 3. Intersection (Multiplication Rule) 4. Independent VS Dependent Events
	2. Counting Principles	1. Counting Principles 2. Product Rule for Counting 3. Simple Applications

	3. Permutations	1. Definition and Concept 2. Permutation Formula P(n,r) 3. Introduction to Factorials
	4. Combinations	1. Definition to Combinations 2. Combination Formula C(n,r) 3. Compare with Permutations 4. Real-life Examples
	5. Applying Permutations and Combinations in Probability	- Probability with Permutations - Probability with Combinations - Problem-Solving Strategy